Magnetic, Electric, and Gravitational Fields

Because magnetic force lines play a big role in our universe and are important for *Interstellar*, let's discuss them, too, before diving into *Interstellar*'s science.

As a student in science class, you may have met magnetic force lines in a beautiful little experiment. Do you remember taking a sheet of paper, placing a bar magnet under it, and sprinkling iron filings (elongated flakes of iron) on top of the paper? The iron filings make the pattern shown in Figure 2.6. They orient themselves along magnetic force lines that otherwise are invisible. The force lines depart from one of the magnet's poles, swing around the magnet, and descend into the other pole. The magnetic *field* is the collection of all the magnetic force lines.

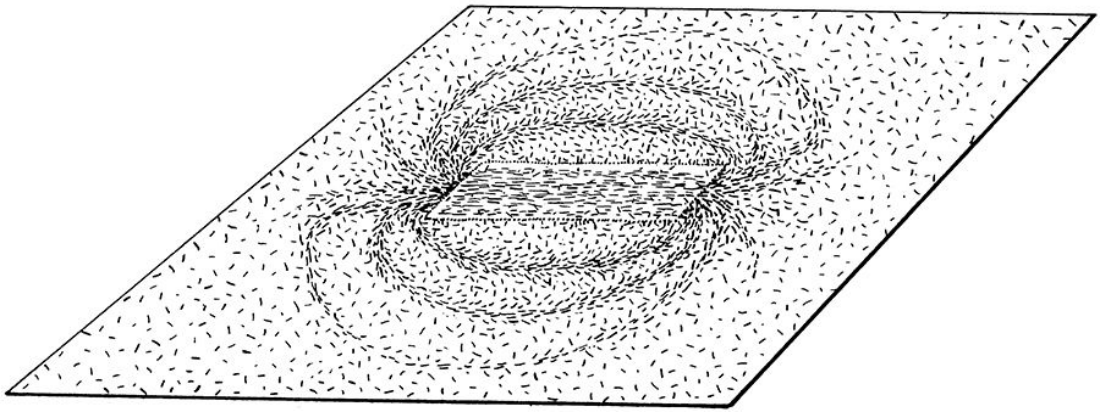


Fig. 2.6. Magnetic force lines from a bar magnet, made visible by iron filings sprinkled on a sheet of paper. *[Drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy.*]*

When you try to push two magnets together with their north poles facing each other, their force lines repel each other. You see nothing between the magnets, but you feel the magnetic field's repulsive force. This can be used for magnetic levitation, suspending a magnetized object—even a railroad train (Figure 2.7)—in midair.